

ARAVINDH R

Software Engineer

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PROFESSIONAL SUMMARY

A passionate and results-driven Software Engineer with 2+ years of professional experience in full stack web development. Proficient in building scalable, high-performance RESTful APIs using Python (FastAPI) and Laravel (PHP), and developing dynamic frontend interfaces with React.js and Next.js. Skilled in designing secure system architectures with JWT and Cookie-based authentication, alongside complex database modeling and query optimization in PostgreSQL and MySQL. Experienced in end-to-end project ownership — from system design and API development to Docker-based containerized deployment on VPS environments with Linux and Nginx configuration, and ongoing production support. Built enterprise-grade solutions including metadata-driven low-code platforms and CRM-based systems for healthcare and inventory domains. Adept at version control using Git and GitHub, ensuring clean workflows and structured release management.

TECHNICAL SKILLS

Languages	C, Python, Java, PHP, JavaScript, SQL
Backend	Python FastAPI, Laravel, REST API Development
Frontend	Next.js, React.js, Redux, HTML5, CSS3, Tailwind CSS, Bootstrap
Automation & Integration	n8n, Meta WhatsApp Webhooks, Razorpay, REST API Integration
Databases	PostgreSQL, MySQL, Alembic Migrations
DevOps	Docker, VPS Deployment (Linux Server Setup), Nginx Configuration
Auth & Security	JWT / Bearer Token Authentication, Cookies
Version Control	Git, GitHub, Collaborative Development, Release Management

WORK EXPERIENCE

Software Engineer

Sep 2024 – Present (2+ years)

Selten Technologies · Full-time | Karaikudi, Tamil Nadu · On-site

- Progressed from Laravel (PHP) to Python (FastAPI) as the primary backend technology, building high-performance, scalable RESTful APIs and production-ready backend systems throughout the role.
- Designed and developed dynamic frontend interfaces using React.js and Next.js, integrating seamlessly with backend APIs for complete end-to-end feature delivery.
- Performed advanced database operations, wrote optimized SQL queries, and managed schema migrations across MySQL and PostgreSQL environments.
- Built secure API architectures with JWT/Bearer token authentication, Cookie-based sessions, third-party integrations, and live VPS server management with Docker-based deployment.

- Engaged in end-to-end project development including system design, coding, testing, deployment, and production support across healthcare and inventory enterprise solutions.

PROJECTS

Sarvam ENT – Hospital Management System

Tech Stack: Next.js, Python FastAPI, PostgreSQL, REST APIs, n8n, Meta WhatsApp Webhooks, Razorpay, JSON Configuration

- Designed and implemented a dynamic module builder allowing users to create custom modules at runtime — eliminating need for code changes when adding new modules or fields.
- Developed a field configuration system supporting multiple field types, validations, and metadata-driven rendering.
- Built dynamic one-to-one, one-to-many, and many-to-many relationships between modules using configurable schemas.
- Implemented layout management system where UI panels and field arrangements are controlled via database-stored JSON configurations.
- Created generic CRUD APIs adapting based on module and field metadata; designed dynamic filtering and querying mechanisms.
- Developed consultation workflow with visit state transitions, auto-generation of consultation PDF reports, and document linking.
- Designed and implemented a WhatsApp healthcare appointment chatbot using n8n and Meta Webhooks, integrating with Selten ONE HMS REST APIs for patient lookup, doctor selection, availability, slot reservation, and appointment booking.
- Implemented state-based conversation management and session persistence using n8n Data Tables, supporting complete appointment workflows with Razorpay Online Payment and Pay at Arrival, including payment-link generation, booking confirmation, and automated notifications.
- Built a low-code / no-code style system within HMS — highly flexible and extensible architecture suitable for enterprise healthcare.

Mahi Diagnostics – Hospital Management System

Tech Stack: React.js, Python FastAPI, PostgreSQL, REST APIs

- Developed patient registration, appointment scheduling, and billing modules using React.js, FastAPI, and PostgreSQL.
- Implemented RBAC-based authentication and automated Cron Jobs for report generation, appointment reminders, and data synchronization.
- Designed and implemented MSSQL-to-PostgreSQL data synchronization pipeline for live data migration and synchronization.

A Square Design Studio – Inventory Management System

Tech Stack: React.js, Redux, Laravel, MySQL, Tailwind CSS, REST APIs

- Developed inventory management system with real-time stock updates and responsive UI using React.js, Redux, Laravel, MySQL, and Tailwind CSS.
- Designed and optimized MySQL database schema for product catalogues, stock movements, and supplier management.
- Developed reporting module with exportable reports, stock history tracking, and analytics dashboard.

EDUCATION

B.Sc. Physics

Raja Doraisingam Government Arts College, Sivaganga | Tamil Nadu, India

2020 – 2023